

- 2 (a) (i) $v^2 = 2as$
 $v^2 = 2 \times 0.85 \times 9.8 \times 12.8$
 $v = 14.6 \text{ m s}^{-1}$ C1
A1 [2]
- (ii) time = $29.3 / 14.6$ C1
 $= 2.0 \text{ s}$ A1 [2]
(any acceleration scores 0 marks; allow 1 s.f.)

- (b) either $60 \text{ km h}^{-1} = 16.7 \text{ m s}^{-1}$
or $14.6 \text{ m s}^{-1} = 53 \text{ km h}^{-1}$
or $22.1 \text{ m s}^{-1} = 79.6 \text{ km h}^{-1}$ M1
so driving within speed limit A1
but reaction time is too long / too slow B1 [3]

- 3 (a) moment: force $\times$ perpendicular distance M1